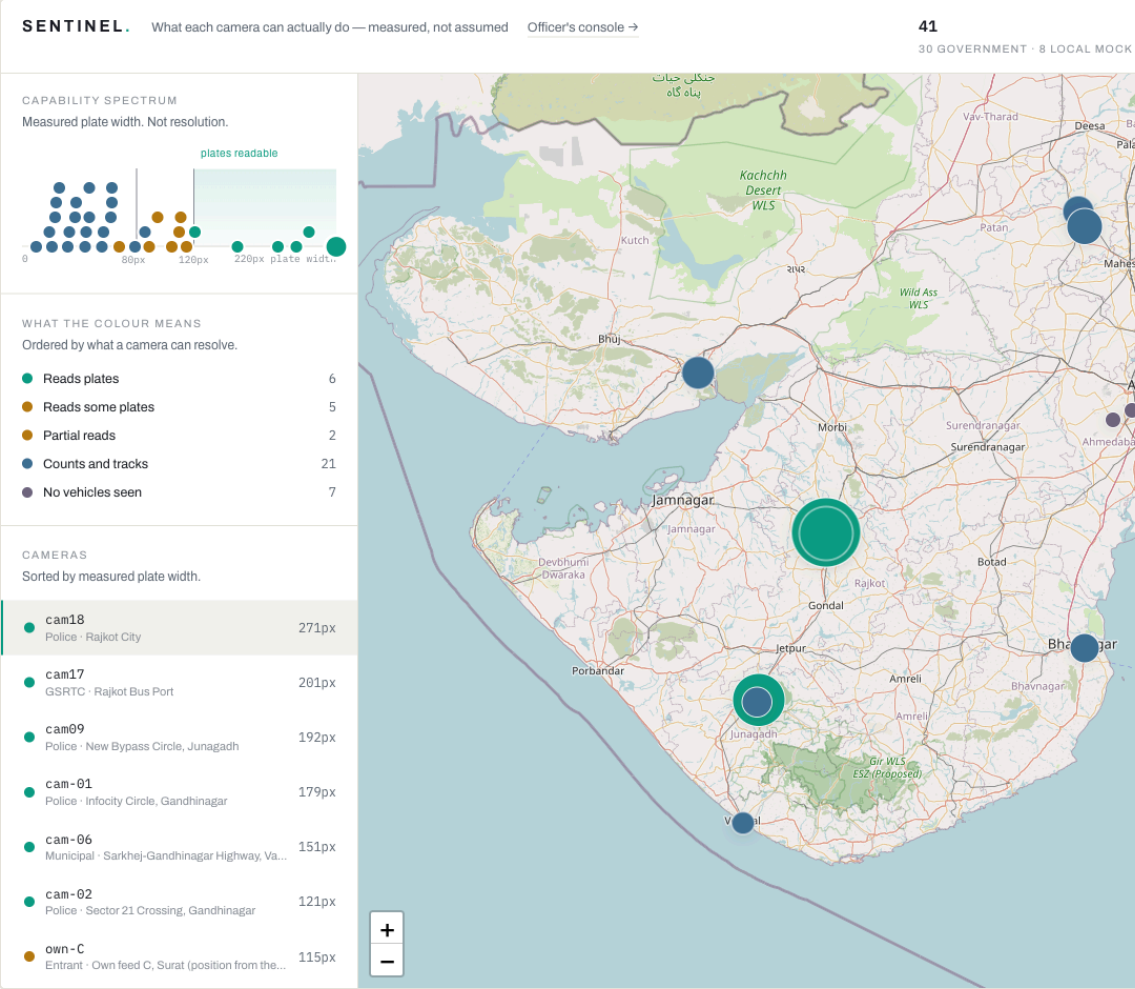


What each camera can actually do.

A camera registry that measures capability, and a control plane that acts on it.



Four things, built and running on your grid today.

- **A registry that measures what each camera can resolve**

Plate width in pixels, clock drift, delivered frame rate, night usability — measured from the camera's own video, never read off a datasheet. 41 cameras carry it now.

- **A control plane that assigns analytics by capability**

Plate reading is scheduled only where a plate is legible. On this grid that is 3 government cameras in 30. The rest do cheap presence work and corroborate.

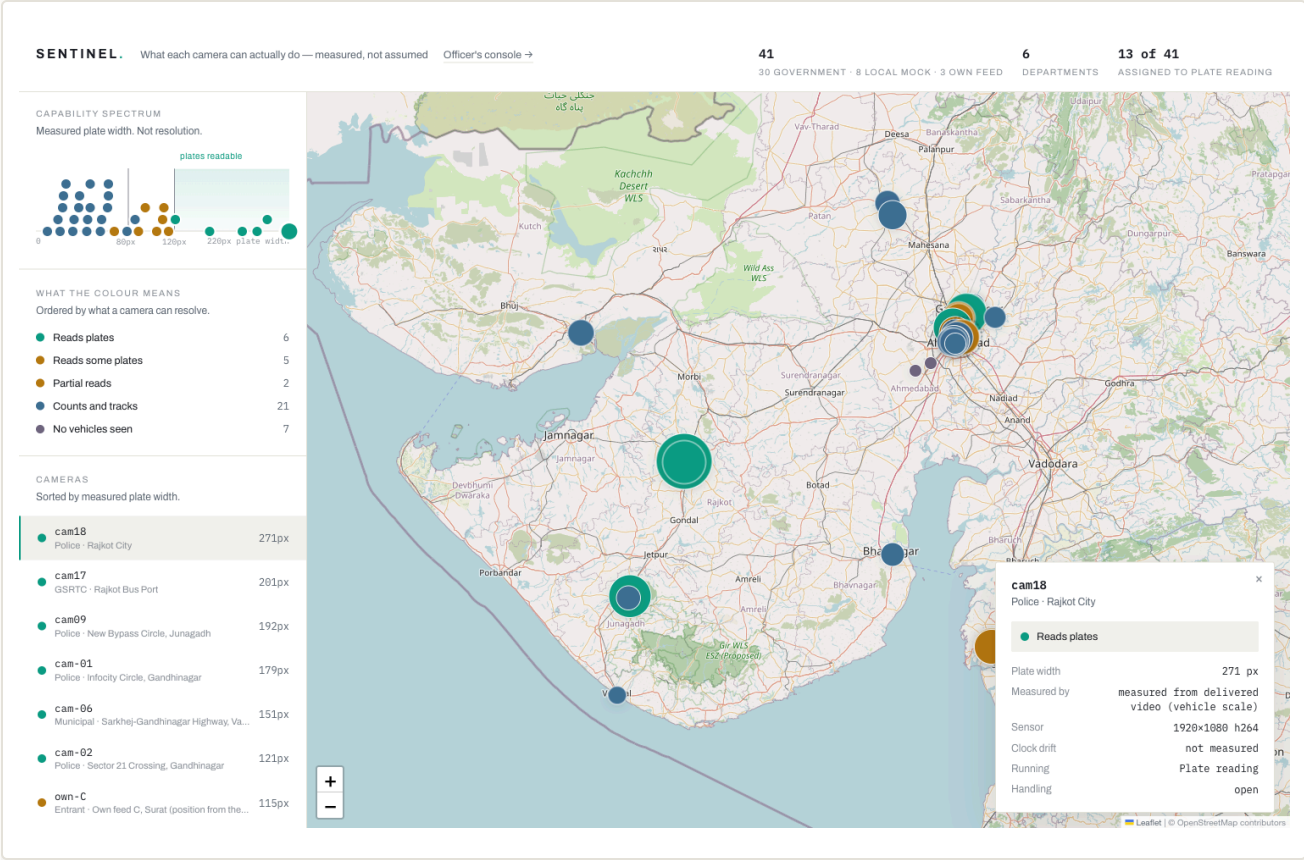
- **An officer's console that shows its working**

Search a registration, get a route with a confidence and a reason for every sighting, the vehicle and the plate crop the reader worked from, and a court-ready bundle in one action.

- **Governance enforced in code, not in a policy annex**

Case-linked access refused before a row is read, a tamper-evident audit chain over every query and export, and a confidence tier capped by what the camera was measured able to do.

A map coloured by what a camera can do, not by whether it is switched on.



41

CAMERAS IN THE REGISTRY

30 government • 8 local mock • 3 own feed

6

DEPARTMENTS FEDERATED

One registry over equipment bought at different times by different owners. No camera is replaced.

428.5s

WORST MEASURED CLOCK DRIFT

Found on a live government camera. Timing is corrected before ordering, and the correction is shown.

Every government camera, on the one axis that decides what it may be asked to do.

Measured plate width p90 pixels across the number plate, from each camera's own delivered video



3 of 30

CAN READ A PLATE

One in 10. Scheduling ANPR on the others would not fail loudly — it would invent registrations.

27.4px

MEDIAN PLATE WIDTH

Against the ~120px a read needs. Resolution does not predict it: 1920×1080 yielded 22px, 854×480 yielded 63px.

2.6%

WORST MEASUREMENT ERROR

The measurement itself was validated against hand-marked plates on 8 cameras, mean 1.02%.

Search a registration. Get a route you can defend.

SENTINEL. Vehicle movement reconstruction — every sighting, with its confidence stated

← CAMERA REGISTRY

6

SIGHTINGS

6

CAMERAS

21.5m

ELAPSED

24.37km

TRACKED

1

IMPOSSIBLE LEGS

VEHICLE SEARCH

Case-linked. Refused without an authorising officer.

REGISTRATION

GJ01AB1234

CASE REFERENCE

FIR-0117/2026

SEARCHED BY

PSI R. Chauhan

AUTHORISED BY

DySP M. Pazmaz

SEARCH

Refused without a case and an authorising officer. The query is written to the **tamper-evident log** before any result is returned.

HUNT LIVE FOOTAGE

Find a designated vehicle on cameras nobody has searched yet. Case-linked to the same authorisation above.

DESIGNATED VEHICLE

Car

Designated in the proven run, from cam09.

THE HUNT DOES NOT RUN HERE

Live hunting runs detection and re-identification over video. That needs PyTorch, OpenCV and a GPU; this is a servelless function with none of them. Everything on this console was computed by the real pipeline offline.

The hunt is shown working end to end on real government footage in the demonstration video.

WATCHLIST

Matched before the officer typed anything.

RECONSTRUCTED MOVEMENT

09 Sept · 6 sightings on 6 cameras across 5 departments · 24.37 km in 21.5 minutes, corrected time order.

IMPOSSIBLE TRANSIT — SIGHTING 5 TO 6

cam-07 to cam-08: 4.873 km in 55 seconds implies **318.9 km/h**. Either two vehicles carry this plate, or a camera clock is wrong. The route below is shown, not corrected.

1

09:11:00 IST · 03:41:00Z

cam-02 · Sector 21 Crossing, Gandhinagar

Police - reads plates LOCAL MOCK

GJ01AB1234 CONFIRMED

94%

NO IMAGE RETAINED FOR THIS SIGHTING

1.924 km 2.8 min 42 km/h

2

09:13:45 IST · 03:43:45Z

cam-01 · Infocity Circle, Gandhinagar

Police - reads plates LOCAL MOCK

GJ01AB1234 PROBABLE

55%

NO IMAGE RETAINED FOR THIS SIGHTING

Read GJ01AB1234, 0.5 from the registration searched — OCR-confusable characters. Recovered by matching, demoted for it.

5.82 km 5.7 min 61.6 km/h

3

09:19:25 IST · 03:49:25Z

cam-05 · Adalaj Village Approach

Panchayat - partial reads LOCAL MOCK

GJ01AB1234 PROBABLE

81%

NO IMAGE RETAINED FOR THIS SIGHTING

cam-05 measures 95px of plate; capped at probable on capability alone.

7.529 km 7.8 min 57.7 km/h

4

09:27:15 IST · 03:57:15Z

cam-06 · Sarkhej-Gandhinagar Highway, Vaishnodedi

Municipal - reads plates LOCAL MOCK

GJ01AB1234 CORROBORATING

44%

NO IMAGE RETAINED FOR THIS SIGHTING

PRODUCE EVIDENCE BUNDLE

6 → 1

SIGHTINGS, ONE ROUTE

Corrected time order across five departments, with the distance and implied speed of every leg shown.

318.9 km/h

IMPOSSIBLE TRANSIT, FLAGGED

Either two vehicles carry this plate or a camera clock is wrong. The route is shown,

not corrected

— an officer is better placed to say which.

3 tiers

NEVER A BOOLEAN

Confirmed, probable, corroborating. A camera that cannot resolve a plate can strengthen a track but can never raise an alert alone.

THE OFFICER'S SCREEN

05 / 12

Every sighting carries the vehicle, and the pixels the reader actually read.



- **Cut from the frame the read came from**

Not the arrival moment. A track is read from its best look, and where the read was fused across several looks the fused image is what is shown — because that is what the reader worked from.

- **Capped at probable, on capability alone**

This camera measures 110px of plate. Whatever the reader's confidence, a camera measured below the threshold cannot produce a confirmed match.

- **Sightings with no image say so**

A row whose footage was not retained reads "no image retained" rather than showing a stand-in.



OWN FEED · SURAT · READ AT 1.00

A bundle that survives being challenged by someone who does not trust me.

Evidence bundle

FIR-0231/2026 · prepared by PSI R. Chauhan · authorised by DySP M. Parmar

Vehicle

GJ19BE3669

Sightings

2

Cameras, with measured capability

2

Impossible legs

0

Audit entries for this case

67

Generated

2026-09-26T19:45:37Z

SHA-256 OVER THE BUNDLE

a1a2c37db0d0390a812ea79303d149e7cf4b75b5044bbb1a6379850cde4d894a

Audit chain: every entry for this case verifies against its recorded predecessor.

DOWNLOAD JSON

COPY HASH

- **Refused before a row is read**
A movement query names the case, the searching officer and the officer authorising it. Without all three it is refused — and the refusal is logged.
- **Tamper-evident by construction**
Every entry hashes the one before it. Editing or deleting one breaks the chain and names the row that broke. 230 entries verify today.
- **The bundle hashes itself into that chain**
SHA-256 over the exported object, written to the log as it leaves. The camera block carries measured capability, so "how do you know it was that vehicle" has a number for an answer.

Facial recognition is assigned by the same control plane — and on this grid it assigns it to nothing.

Face matching is limited by inter-ocular distance. ISO/IEC 19794-5 puts **60px** as the enrolment floor; I use a generous **40px** to even attempt one, derived from person detections through published anthropometry so a reviewer can recompute it. The console carries it per camera, beside plate width — the same question, asked of people.

9.8px

BEST CAMERA MEASURED

Short by a factor of 4.1. The worst gives 2.7px — not a face, a smudge that resembles one.

0

CAMERAS ASSIGNED FACE MATCHING

Measured across 1,737 people on 4 cameras. Onboard one that clears the floor and it is assigned automatically, capped at probable.

3 layers

PERSON SAFETY WITHOUT A WEAPON DETECTOR

A knife is a 2px smear here. Twenty people running from one point is a third of the frame — that is what is detected. False alarms measured on real government footage; detection measured on simulation.

Clock drift	not measured
Running	Plate reading
Handling	open

FOR PEOPLE

Between the eyes	9.8 px · 40px needed to attempt, 60px to enrol — short by 4.1x
Face matching	No face matching here
Body height	370 px · from 139 people measured
Threat posture	Gate open · 370px of body against a 160px floor; limb angles are noise below it. Not validated — no real footage of the behaviour exists, so the layer is not run.
Weapon in hand	Gate open · a hand here is 22.2px against a 8px floor. Not validated — not enabled on any camera.
Crowd dispersal	Available · reads the crowd's motion field, so it needs neither a face nor a weapon. False alarms measured on real government footage, day and night; detection measured on simulated crowds only.

Facial recognition is assigned by the same control plane that assigns plate reading — and it assigns none to this camera. At 9.8px between the eyes there is no face present, only a smudge that resembles one.

At **80,000 cameras** the registry stops being a nicety and becomes the budget.

	RUN ANALYTICS ON EVERY CAMERA	ASSIGN BY MEASURED CAPABILITY
Cameras on heavy inference	80,000	8,000
Datacentre GPUs required	~1,600	~160
Cameras on cheap presence work	0	72,000
Raw video if centralised	160-400 Gbps sustained	2-5% of raw – metadata and evidence clips leave the edge
Storage at full retention	103 PB at 30 days	72h hot • 1yr warm • 7yr case-bound

The gap between the two columns is the case for the whole design. Nothing here is a projection from a vendor sheet: the unit cost comes from measured throughput, and the capability share from the 30-camera survey.

The six scalability requirements, each with a measured answer.

REQUIREMENT	ANSWER	THE NUMBER BEHIND IT
Hardware & software	Edge, district and central tiers on commodity hardware. Permissive licences only — no AGPL in the inference path.	Apache-2.0 / BSD-3
AI processing capacity	50 streams per datacentre GPU as the planning midpoint, applied to the assigned subset rather than the estate.	~160 GPUs, not ~1,600
Network & bandwidth	Inference at the edge; metadata and evidence clips cross the network, not raw video. Low-bandwidth sites degrade to sampled frames rather than dropping out.	95–98% reduction
Storage & retention	Hot / warm / cold tiers, per-department retention driven by the registry's governance class.	72h · 1yr · 7yr
Disaster recovery	Cameras re-home on edge failure; standby district; warm central standby with streaming replication of the registry.	RT0 < 1h district RPO < 5 min
Statewide rollout	Six phases from the sandbox proof to 33 districts, each with a stated exit criterion — including a lens-and-aim phase before expanding.	18 months to 80,000

Underclaimed on purpose. The next stage is live feeds in front of you.

RUNNING TODAY

- **Measured on the real grid**

30 of 30 government cameras surveyed, 0 failures, capability and clock drift recorded per camera.

- **End to end on live feeds**

Ingest forced over TCP, timing from PTS, reconnect with backoff — demonstrated on government cameras and on the entrant's own footage.

- **Tested**

210 automated tests pass, including the organisers' own pre-submission checklist as executable acceptance tests.

DESIGNED, NOT YET BUILT

- **Authentication and RBAC**

The case-linked gate is real and tested; the identity behind it belongs to eGujCop or departmental AD with MFA. Stated plainly rather than implied.

- **Masking on export**

Uninvolved parties should be obscured in bundles. The hosted build withholds them; obscuring them inside an image a case is entitled to is not built.

- **Person re-identification and dispersal**

Validated in simulation only. There is no government footage of an attack, and there will not be one.

Next: the live test on 12–13 October — ~50 heterogeneous cameras onboarded, a designated vehicle tracked, alerts, movement history and GIS, on the infrastructure the state already owns.

Thank you.

Open the console and try it. Search a registration, clear the case reference, and watch it refuse.

HOSTED CONSOLE

`sentinel-gujaratpolice-demo.vercel.app`

COMMITTEE CREDENTIALS

`reviewer / gujarat2026`

SOURCE

`github.com/meherzankturel/sentinel-gujarat-police`

ENTRANT

`Meherzan Turel · individual`